



MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

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Chair of Empirical Microeconomics

ADVANCED MICROECONOMICS

Consumer Theory IV + V - Demand Functions and Duality

Winter Term 2025/2026

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Literature

- Autor (2016): Lecture Notes 6 + 7
- **Application:** Jensen, R.T., & N.H. Miller (2008): Giffen Behavior and Subsistence Consumption. AER 98 (4): 1553–77.

Overview

1 Duality and Demand functions

Duality

Marshallian demand

Hicksian demand

2 Shephard's Lemma, Roy's Identity and Slutsky equation

Compensated and uncompensated demand

Shephard's Lemma

Slutsky Equation

Roy's Identity

3 Comparative Statics: Effects of price changes

4 Types of Goods

5 Application: Do Giffen goods exist in reality?

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Recap: Approaches

1: Utility maximization (primal approach)

$$\max_{x,y \geq 0} U(x,y) \text{ s.t. } p_x x + p_y y \leq I$$

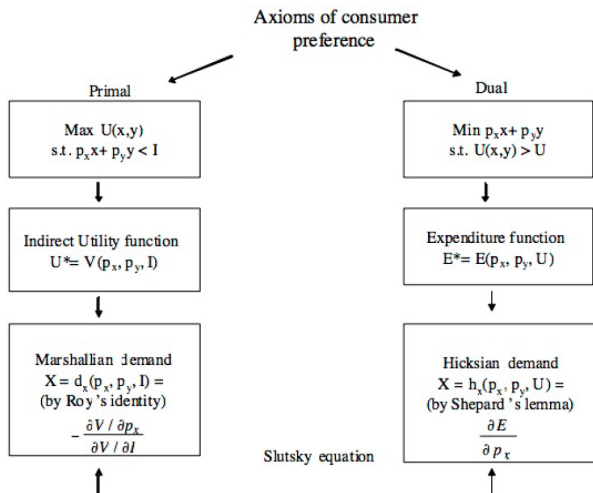
Solution delivers the optimal amounts x^*, y^* (as functions of p_x, p_y , and I), plugging them into the utility function gives $u^* = U(x^*, y^*)$ for given p_x, p_y, I . In general, we get the **indirect utility function** $V(p_x, p_y, I) = u^*$.

2: Expenditure minimization (dual approach)

$$\min_{x,y \geq 0} p_x x + p_y y \text{ s.t. } U(x,y) \geq u^*$$

Solution delivers optimal amounts x^*, y^* (as functions of p_x, p_y , and u^*), plugging into the budget restriction (*cost function*) gives the **expenditure function**: $E(p_x, p_y, u^*) = p_x x^* + p_y y^*$.

Duality



Source: Autor (2016), Lecture Notes 4, p. 2

Duality of expenditure and indirect utility function

Problem	Given	Unknown
Utility Maximization	Income I_0	Maximal Utility $U_0 = V(p_x, p_y, I_0)$ (a)
Expenditure Minimization	Utility U_0	Minimal Expenditures $I_0 = E(p_x, p_y, U_0)$ (b)

- Indirect utility function and the expenditure functions are **inverse** functions:

$$V(p_x, p_y, E(p_x, p_y, U_0)) = U_0 \quad (\text{Plugging } b \text{ into } a)$$

and

$$E(p_x, p_y, V(p_x, p_y, I_0)) = I_0 \quad (\text{Plugging } a \text{ into } b)$$

- Both problems deliver the same optimal consumption bundle x^*, y^*

Duality in our example

- Utility function $U(x, y) = x^{\frac{1}{2}} y^{\frac{1}{2}}$
- Indirect Utility function:

$$V(p_x, p_y, I) = \left(\frac{I}{2p_x} \right)^{\frac{1}{2}} \left(\frac{I}{2p_y} \right)^{\frac{1}{2}}.$$

- Plugging this into the expenditure function (*lecture 3, slide 15*) $E(p_x, p_y, U_p) = 2p_x^{\frac{1}{2}} p_y^{\frac{1}{2}} U_p$ results in

$$\begin{aligned} E(p_x, p_y, U_p) &= E(p_x, p_y, V(p_x, p_y, I_p)) \\ &= 2p_x^{\frac{1}{2}} p_y^{\frac{1}{2}} \left(\frac{I_p}{2p_x} \right)^{\frac{1}{2}} \left(\frac{I_p}{2p_y} \right)^{\frac{1}{2}} \\ &= I_p. \end{aligned}$$

Demand functions

- Constrained utility maximization $\Rightarrow$ indirect utility function
- Expenditure minimization $\Rightarrow$ expenditure function
- Implicitly, we have seen the **demand functions**: the optimal **amounts demanded** of the different goods **depending on prices and ...**
 - income (utility max approach)
 - utility (cost min approach)
- These are the **Marshallian** and **Hicksian** demand functions.

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Marshallian demand

Marshallian Demand

gives **demanded quantity** of a good as a function of **prices** and **income**

- Goes back to Alfred Marshall (British economist, 1842–1924).
- Mathematically, it results from the **constrained utility maximization approach**.
- Notation of the **Marshallian demand of good x** :

$$d_x(p_x, p_y, I),$$

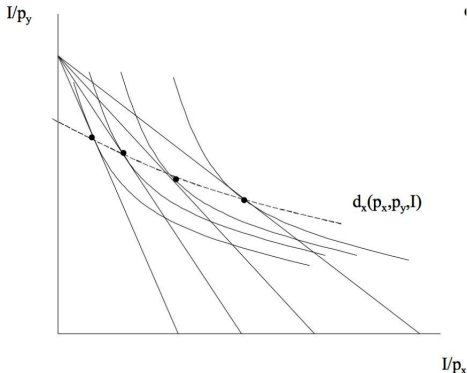
for the case of goods x and y with prices p_x , p_y and income I .

- Allows to see how demand changes with changes in the price of one good, ceteris paribus.
- **Marshallian demand** also called **uncompensated demand** and measures the **total effect** of a price change.

Marshallian demand

Graphically

- Set of points of tangency between indifferent curves and the **budget lines that tilts with price changes** (*here: decrease of p_x or increase of p_y*)



Marshallian demand

Mathematically

- Goal: Derivation of optimal amounts of x and y with given I, p_x, p_y
- Example: $U(x, y) = x^{\frac{1}{2}} y^{\frac{1}{2}}$
- The Lagrange-Approach to solve the **constrained utility maximization** problem (*Lecture 2*) resulted in:

$$x^* = d_x(p_x, p_y, I) = \frac{I}{2p_x}$$

and

$$y^* = d_y(p_x, p_y, I) = \frac{I}{2p_y}$$

→ These are the **Marshallian demand functions**.

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Hicksian demand

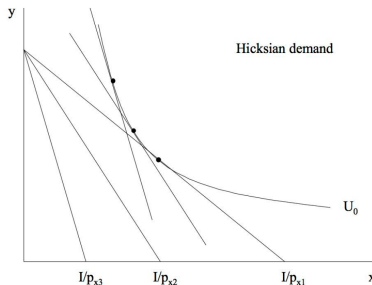
Hicksian Demand

measures demanded quantities as function of prices and **utility**

- Goes back to British economist John Richard Hicks (1904–1989) who received the Nobel prize in economics in 1972.
- Results from expenditure minimization approach.
- Thus it measures how demand changes with a change in price holding the prices of all other goods and utility constant.
- Hicksian demand also called **compensated demand** (*as change in real income is compensated for*).

Hicksian demand

- **Hicksian demand:** Effect of a price change while holding utility level constant, i.e. compensating change of relative income.



Source: Autor (2016), Lecture Notes 6, p. 5

- **Graphically:** Set of points of tangency between one (fixed) indifference curve and different budget lines that tilt due to the price change and shift so that the fixed indifference curve can be reached despite the price change.

Hicksian demand

- **Example** (*continued*): $U(x, y) = x^{\frac{1}{2}} y^{\frac{1}{2}}$.
- The Lagrange-Approach for expenditure minimization resulted in the optimal demanded amounts depending on **prices** and **utility level**:

$$x^* = h_x(p_x, p_y, U) = \left(\frac{p_y}{p_x} \right)^{\frac{1}{2}} U$$

and

$$y^* = h_y(p_x, p_y, U) = \left(\frac{p_x}{p_y} \right)^{\frac{1}{2}} U$$

- These are the **Hicksian demand functions**.

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Duality - Relation between the demand functions

- The demand functions tell us how optimal demanded quantities of goods react to price changes, **ceteris paribus**.
- Ceteris paribus for **Marshallian (uncompensated)** demand refers to prices of other goods **and income**.
 - Marshallian demand measures **total effect** of price change.
- Ceteris paribus for **Hicksian (compensated) demand** refers to prices of other goods **and utility**.
 - Hicksian demand only measures the effect of the price change when utility is hold constant, i.e. change in relative income is compensated for.

Duality - Relation between the demand functions

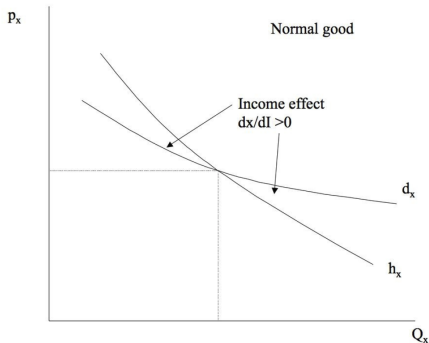
- Mathematically, the duality of the two optimization approaches (utility maximization and expenditure minimization) implies the duality of the resulting demand functions $h_x(p_x, p_y, \bar{U})$ and $d_x(p_x, p_y, I)$ in the optimum:

$$h_x(p_x, p_y, \bar{U}) = d_x(p_x, p_y, E(p_x, p_y, \bar{U}))$$

- For every fixed utility level ($\bar{U}$) Hicksian (h_x) equal to Marshallian demand (d_x) if income necessary to reach $\bar{U}$ at given prices p_x, p_y ($E(p_x, p_y, \bar{U})$) is given.

Relation between the demand functions

Graphical illustration (with normal good)



Source: Autor (2016), LN 7, p. 8

- Demand functions intersect at starting point $p_x, p_y, \bar{U}$ and $I = E(p_x, p_y, \bar{U})$.
- Because they measure different reactions to price changes demand curves look different in general.
- The exact shape of the demand curves depends on the **type of the good**.

Duality – Relationship between the Demand Functions

Mathematical

- Mathematically, from $h_x(p_x, p_y, \bar{U}) = d_x(p_x, p_y, E(p_x, p_y, \bar{U}))$ it follows that:

$$\frac{\partial h_x}{\partial p_x} = \frac{\partial d_x}{\partial p_x} + \frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x}$$

$$\frac{\partial d_x}{\partial p_x} = \frac{\partial h_x}{\partial p_x} - \frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x}$$

- Derivatives correspond to the different slopes of the demand curves
- Change in uncompensated demand** $(\frac{\partial d_x}{\partial p_x})$ thus corresponds to the **change in compensated demand** $(\frac{\partial h_x}{\partial p_x})$ minus $\frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x}$.

Relation between the demand functions

$$\frac{\partial d_x}{\partial p_x} = \frac{\partial h_x}{\partial p_x} - \frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x}$$

- What does $\frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x}$ mean?
- In general: Effect of the income change (necessary to compensate for utility change) on demand
- Consisting of:
 - 1 $\frac{\partial d_x}{\partial I}$ is the change in demand of x if income I change by small amount.
 - 2 $\frac{\partial E}{\partial p_x}$ describes how strong the necessary income for constant utility changes if prices changes
- We will see that $\frac{\partial E}{\partial p_x}$ is the cost-minimizing quantity optimal demanded of x at $\bar{U}$ and p_x, p_y .

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Shephard's Lemma

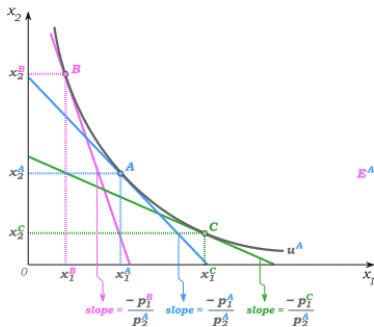
Shephard's Lemma

The derivative of the expenditure function with respect to the price of a good equals the Hicksian demand function for that good:

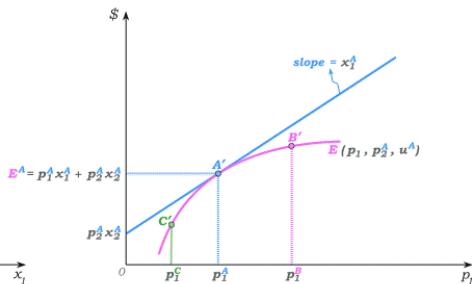
$$\frac{\partial E}{\partial p_x} = h_x = x^*$$

- **Intuition:** The additional expenditure required for a **small (infinitesimal) increase in the price** of good x , such that the consumer can still achieve the utility level $\bar{U}$, corresponds exactly to the demand quantity of x .
- In Shephard's Lemma, we consider a very small (infinitesimal) price change, so that at the optimum there is **no re-optimization** (*the consumer remains passive*), and the necessary expenditure therefore increases exactly by the amount demanded.

Shephard's Lemma



Quelle: Nachyba - Microeconomics: An Intuitive Approach with Calculus.



- **Example:** If you buy two cups of coffee at the Mensa every day and the price increases by 1 cent, you'll need 2 additional cents to keep the utility level constant ($\bar{U}$)
- In case of a larger price change you would **re-optimize**, buy less coffee. In this case, less than 2 cents would suffice to keep $\bar{U}$.

Shephard's Lemma

Derivation

- This in line with the **Envelope Theorem**
 - Recap: Envelope Theorem says that at the optimum in case of small changes only the **direct effect** of the change is relevant, **indirect effects** resulting from re-optimization of other variables can be ignored.
- Mathematical derivation can show this:
 - $E(p_x, p_y, \bar{U})$ is expenditure function from cost-minimization approach:

$$\min_{x,y} p_x x + p_y y \text{ s.t. } U(x, y) \geq \bar{U}$$

- This resulted in Lagrange function

$$\mathcal{L} = p_x x + p_y y + \lambda(\bar{U} - U(x, y))$$

- $E()$ is value function of this approach (the optimum)

Shephard's Lemma

Derivation

- We can find $\frac{\partial E}{\partial p_x}$ by taking total derivative of $\mathcal{L}$ w.r.t. p_x and **evaluate at the optimum**, i.e., at x^*, y^* :

$$\frac{d\mathcal{L}}{dp_x} = \frac{\partial \mathcal{L}}{\partial p_x} + \frac{\partial \mathcal{L}}{\partial x} \frac{\partial x}{\partial p_x} + \frac{\partial \mathcal{L}}{\partial y} \frac{\partial y}{\partial p_x}$$

$$\frac{d\mathcal{L}}{dp_x} = x + \left(p_x \frac{\partial x}{\partial p_x} - \lambda \frac{\partial U}{\partial x} \frac{\partial x}{\partial p_x} \right) + \left(p_y \frac{\partial y}{\partial p_x} - \lambda \frac{\partial U}{\partial y} \frac{\partial y}{\partial p_x} \right)$$

- The FOCs yield $\lambda = \frac{p_x}{\partial U / \partial x} = \frac{p_y}{\partial U / \partial y}$.

Shephard's Lemma

Derivation

- Plugging in λ and evaluate at optimum:

$$\begin{aligned} \frac{\partial E}{\partial p_x} &= \frac{d\mathcal{L}}{dp_x} \Big|_{x^*, y^*} = x^* + \left(p_x \frac{\partial x^*}{\partial p_x} - \frac{p_x}{\partial U / \partial x} \frac{\partial U}{\partial x} \frac{\partial x^*}{\partial p_x} \right) \\ &\quad + \left(p_y \frac{\partial y^*}{\partial p_x} - \frac{p_y}{\partial U / \partial y} \frac{\partial U}{\partial y} \frac{\partial y^*}{\partial p_x} \right) \\ &= x^* = h_x \end{aligned}$$

- Last equality holds because $E()$ holds utility constant, and any resulting demand function must also hold utility constant.
- At the optimum, the indirect effects of x^* and y^* cancel out — this is the essence of the envelope theorem.

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Shephard's Lemma and the Slutsky Equation

- When we plug **Shephard's Lemma** into the **duality relationship** (*from slide 25*), we obtain the **Slutsky Equation**:

Slutsky Equation

$$\begin{aligned}\frac{\partial d_x}{\partial p_x} &= \frac{\partial h_x}{\partial p_x} - \frac{\partial d_x}{\partial I} \frac{\partial E}{\partial p_x} \\ &= \frac{\partial h_x}{\partial p_x} - \frac{\partial d_x}{\partial I} x\end{aligned}$$

Economic interpretation of Slutsky equation

- **Interpretation:** The **total effect** of a price change $\left(\frac{\partial d_x}{\partial p_x}\right)$ is equal to
 - the compensated demand effect $\left(\frac{\partial h_x}{\partial p_x}\right)$
 - minus **income effect** stemming from price change which has two parts:
 - ① change in demand for x following small change in income $\left(\frac{\partial d_x}{\partial I}\right)$, that is
 - ② scaled with the quantity x
- Income effect (following a price change) depends on the quantity demanded before the price change
 - if $x = 0$ income effect doesn't play a role.
 - if x very large, income effect is ceteris paribus large.

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Roy's Identity

- Dual to the situation of compensated demand and the expenditure function (Shephard's Lemma), we can also derive uncompensated demand based on the indirect utility function:

Roy's Identity

$$-\frac{\partial V / \partial p_x}{\partial V / \partial I} = d_x$$

- **Intuition:** The uncompensated (Marshallian) demand for a good is given by the ratio of how strongly utility responds to price changes relative to how strongly it responds to income changes.

Roy's Identity

Derivation 1

- Roy's Identity can be derived from Lagrange function and plugging in optimality conditions:

- Indirect utility function**

$$V = \max_{x,y} U(x,y) \text{ s.t. } p_x x + p_y y \leq I$$

- Lagrange function** $\mathcal{L} = U(x,y) + \lambda(I - p_x x - p_y y)$

- Derivative w.r.t. p_x :**

$$\frac{\partial \mathcal{L}}{\partial p_x} = -\lambda x + \left(\frac{\partial U}{\partial x} \frac{\partial x}{\partial p_x} - \lambda p_x \frac{\partial x}{\partial p_x} \right) + \left(\frac{\partial U}{\partial y} \frac{\partial y}{\partial p_x} - \lambda p_y \frac{\partial y}{\partial p_x} \right)$$

Roy's Identity

Derivation 2

- Based on FOCs we know that in optimum $\lambda = \frac{\frac{\partial U}{\partial x}}{p_x} = \frac{\frac{\partial U}{\partial y}}{p_y}$ (see lecture 3, slide 19).
- Thus we can plug in $\frac{\partial U}{\partial x} = \lambda p_x$ and $\frac{\partial U}{\partial y} = \lambda p_y$. This yields

$$\frac{\partial V}{\partial p_x} = \frac{\partial \mathcal{L}}{\partial p_x} \Big|_{x=x^*, y=y^*} = -\lambda x^*.$$

- How to get rid off λ ?
 - In lecture 3 we have shown that $\lambda = \frac{\partial \mathcal{L}}{\partial I} \Big|_{x=x^*, y=y^*}$ (the shadow price) or $\lambda = \frac{\partial V}{\partial I}$.
- Thus **Roy's Identity** is

$$-\frac{\partial V / \partial p_x}{\partial V / \partial I} = -\frac{-\lambda x^*}{\lambda} = x^* = d_x.$$

- where x^* is optimal x based on utility maximization approach, thus d_x .

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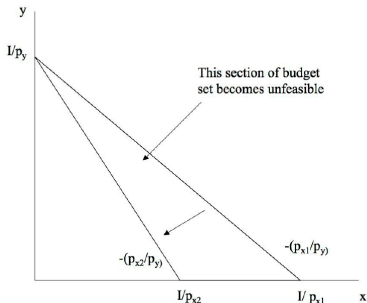
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Effect of a price change

Graphical illustration



Impact of an Increase in P_x on the Budget Set

Source: Autor (2016), Lecture Notes 6, p. 7

- If the price of one good changes (*ceteris paribus*), the budget line tilts
- This has two effects:
 - 1 the budget set changes (*change in real income*)
 - 2 the slope of the budget line changes (*change in relative price ratio*)

Marshallian and Hicksian demand

- Marshallian demand measures the **total effect**.
- Total effect can be split in

Income Effect (IE)

How does demand for x change due to the relative change in real income cause by the price change (*i.e. reaching a higher/lower utility level*)?

Substitution Effect (SE)

How does demand for x change if price for x changes relative to y (=all other goods), but **utility is constant**.

- Hicksian demand measures the **substitution effect**.

Effects of a Price Change

- **Example:** Price increase for good x :

	Uncomp. Demand (Marshallian)	Comp. Demand (Hicksian)
Consumption of x	SE: $-$, IE: $+/-$	SE: $-$, IE: 0
Consumption of y	SE: $+$, IE: $+/-$	SE: $+$, IE: 0
Utility	$-$	0

Substitution effect

Hicksian Decomposition (of total effect)

SE is measured by compensating the consumer for the relative change in income such that she can hold her **utility constant**:

$$\frac{\partial x^*}{\partial p_x} \Big|_{U=U_0}$$

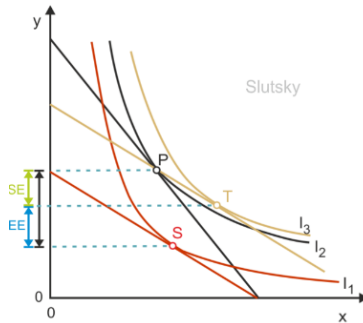
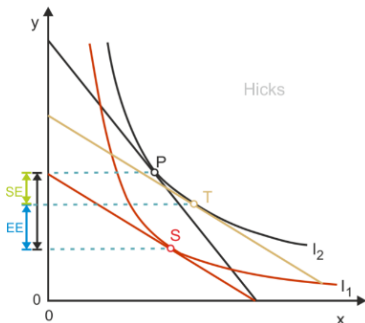
Slutsky Decomposition (of total effect)

SE is measured by compensating the consumer for relative income change such that she can still afford the **same bundle of goods**:

$$\frac{\partial x^*}{\partial p_x} \Big|_{x=x^*, y=y^*}$$

Hicks vs. Slutsky

- **Substitution effects** differ but whatever decomposition we use, it is always **negative**.



Quelle: <https://mikrooekonomie.de/>

- *We will concentrate on the Hicksian decomposition from here on.*

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Types of Goods and Income effect

- In general: Income effect can be **positive or negative**.
 - 1 **Normal Good** - Consumption increases with increasing income:

$$\frac{\partial x^*}{\partial I} > 0$$

- For normal goods income and substitution effect have the same sign.
- 2 **Inferior Goods** - Consumption decreases with increasing income:

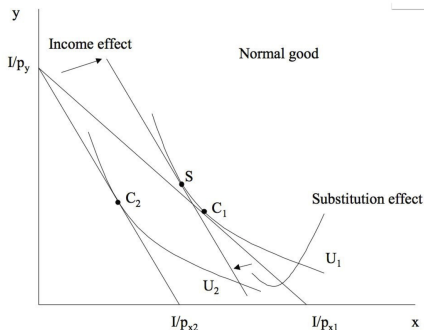
$$\frac{\partial x^*}{\partial I} < 0$$

- For inferior goods income and substitution effect have opposite signs, so that the sign of the total effect depends on the size of the two effects.

Types of goods - Normal Goods

$$\frac{\partial x^*}{\partial I} > 0$$

$$\frac{\partial x^*}{\partial p_x} \Big|_{U=U_0} < 0$$



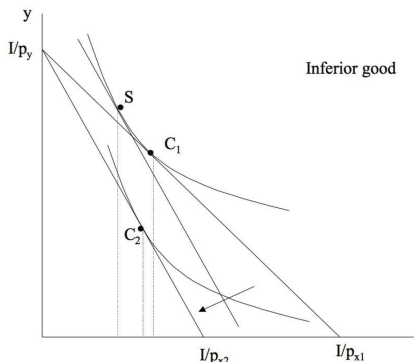
Source: Autor (2016), LN 7, p.2

- Income ($x_{C_2} - x_S$) and substitution effect ($x_S - x_{C_1}$) work in same direction
- Demand decreases with increase in price ($x_{C_2} - x_{C_1} < 0$)

Types of goods - (Weakly) inferior goods

$$\frac{\partial x^*}{\partial I} < 0$$

$$\frac{\partial x^*}{\partial p_x} \Big|_{U=U_0} < 0$$



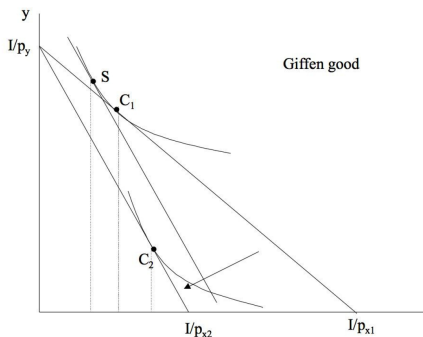
Source: Autor (2016), LN 7, p.3

- Income ($x_{C_2} - x_S$) and substitution effect ($x_S - x_{C_1}$) **have opposite signs**, BUT the **substitution effect is stronger**
- Demand decreases with increase in price ($x_{C_2} - x_{C_1} < 0$)

Types of goods - (Strongly) inferior goods / Giffen goods

$$\frac{\partial x^*}{\partial I} < 0$$

$$\frac{\partial x^*}{\partial p_x} \big|_{U=U_0} < 0$$



Source: Autor (2016), LN 7, p.4

- income ($x_{C_2} - x_S$) and substitution effect ($x_S - x_{C_1}$) **have opposite signs, BUT income effect is stronger**
- **Demand increases with increase price**
($x_{C_2} - x_{C_1} > 0$)

Slutsky Equation and Types of Goods

Slutsky Equation

$$\frac{\partial d_x}{\partial p_x} = \underbrace{\frac{\partial h_x}{\partial p_x}}_{\text{Substitution effect}} - \underbrace{\frac{\partial d_x}{\partial I} x}_{\text{Income effect}}$$

Good Type	$\frac{\partial d_x}{\partial p_x}$	$\frac{\partial h_x}{\partial p_x}$	$\frac{\partial d_x}{\partial I} x$
Normal good	< 0	< 0	> 0
Inferior good			
Weakly	< 0	< 0	< 0 but $ \frac{\partial h_x}{\partial p_x} > \frac{\partial d_x}{\partial I} x $
Strongly (Giffen)	> 0	< 0	< 0 and $ \frac{\partial h_x}{\partial p_x} < \frac{\partial d_x}{\partial I} x $

Overview

1 Duality and Demand functions

Duality

Marshallian demand

Hicksian demand

2 Shephard's Lemma, Roy's Identity and Slutsky equation

Compensated and uncompensated demand

Shephard's Lemma

Slutsky Equation

Roy's Identity

3 Comparative Statics: Effects of price changes

4 Types of Goods

5 Application: Do Giffen goods exist in reality?

Jensen and Miller, 2008

American Economic Review 2008, 98:4, 1553–1577
<http://www.aeaweb.org/articles.php?doi=10.1257/aer.98.4.1553>

Giffen Behavior and Subsistence Consumption

By ROBERT T. JENSEN AND NOLAN H. MILLER*

This paper provides the first real-world evidence of Giffen behavior, i.e., upward sloping demand. Subsidizing the prices of dietary staples for extremely poor households in two provinces of China, we find strong evidence of Giffen behavior for rice in Hunan, and weaker evidence for wheat in Gansu. The data provide new insight into the consumption behavior of the poor, who act as though maximizing utility subject to subsistence concerns. We find that their elasticity of demand depends significantly, and nonlinearly, on the severity of their poverty. Understanding this heterogeneity is important for the effective design of welfare programs for the poor. (JEL D12, O12)

Questions about the study

- We will discuss the following questions:
 - What is the paper about?
 - How is it related to our class?
 - What data and methods are used?
 - What are the paper's (main) results?
 - How valid are the results?
 - What is your personal take-away from this paper?

Summary

- ① **Duality:** Utility maximization $\leftrightarrow$ Expenditure minimization $\Rightarrow$ Marshallian (d_x) and Hicksian (h_x) demand.

② **Key Relationships**

- Shephard's Lemma: $\frac{\partial E}{\partial p_x} = h_x$
- Roy's Identity: $-\frac{\partial V / \partial p_x}{\partial V / \partial I} = d_x$
- Slutsky Equation: $\frac{\partial d_x}{\partial p_x} = \frac{\partial h_x}{\partial p_x} - \frac{\partial d_x}{\partial I} x$

③ **Interpretation:**

- Hicksian demand isolates the *substitution effect*.
- Marshallian demand includes *income and substitution effects*.

④ **Types of Goods:**

- Normal: $\frac{\partial d_x}{\partial I} > 0$, $\frac{\partial d_x}{\partial p_x} < 0$
- Inferior (weakly): $\frac{\partial d_x}{\partial I} < 0$, $|\text{SE}| > |\text{IE}|$
- Giffen: $\frac{\partial d_x}{\partial I} < 0$, $|\text{SE}| < |\text{IE}|$

Important Keywords

- ➊ Duality of consumer choice
- ➋ Marshallian (uncompensated) demand
- ➌ Hicksian (compensated) demand
- ➍ Shephard's Lemma
- ➎ Roy's Identity
- ➏ Slutsky Equation
- ➐ Substitution effect vs. Income effect
- ➑ Normal, Inferior, and Giffen goods

Fragen?

